

FeatDAE: Introducing Features With Denoising Autoencoder for Anomaly Detection

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Abstract—Anomaly detection (AD) is a critical task in manufacturing inspection. Reconstructive AD methods restore the normal appearance of an object, ideally modifying only the anomalous regions. However, previously commonly used reconstruction-based architecture always struggles with over-generalization and overfitting problems, leading to poor reconstruction performance on real defective samples. In this study, we propose a more general denoising autoencoder, by introducing a feature hierarchy design to address these challenges in unsupervised AD. In particular, we operate feature transformation in the latent space to cope with the robustness of unseen anomalies in reality. Furthermore, the method enhances the discriminative capability of the model by focusing on multiple knowledge, including pixel color, histogram of oriented gradient (HOG) feature, and deep features. Additionally, a feature alignment module is proposed to manage the varied sizes and morphologies of features. Experiments conducted on both the MVTec AD dataset and the VisA dataset demonstrate that our FeatDAE significantly outperforms existing methods, achieving state-of-the-art results with high efficiency.

Index Terms—Anomaly detection (AD), denoising autoencoder, industrial defect detection, self-supervised learning, unsupervised learning.

I. INTRODUCTION

ANOMALY detection (AD) aims to identify instances that contain anomalous and localized regions that deviate from normal object patterns, playing an essential role in various applications such as industrial quality control [1], [2], [3]. However, the scarcity and diversity of anomalies pose significant challenges in collecting and labeling, resulting in the inability to construct large datasets to train a classifier or detector. In addition, the pattern differences between normal and abnormal images are usually fine-grained, while defects may occupy only a small fraction of image pixels. Therefore, AD based on only normal data encounters unique challenges different from those of a typical supervised learning problem.

Many recently popular and successful unsupervised AD models are reconstruction-based [4], [5]. These models learn to

construct a latent space of normal samples by minimizing the discrepancy between the original normal instances and their reconstructions. It can be further subdivided into the following two paradigms. **Generative** methods [6], [7], [8] assume that an autoencoder will be optimized to reproduce normal patterns, thereby failing to accurately reconstruct anomalous regions. As a result, the poor reconstruction of defective areas leads to significant differences between the reconstructed output and the input images. This discrepancy can be used to identify and localize the anomalous regions. However, the high generalization capacity of autoencoders allows them to even reconstruct anomaly fidelity, especially when similar local features are presented [9], as illustrated in Fig. 1(a). This overgeneralization violates the basic assumption of generating anomaly maps based solely on the reconstruction error, significantly increasing the rate of false negatives (FNs). **Discriminative** methods [1], [10], [11] train models by synthesizing anomalies to extend these models to recognize real defects. They simultaneously generate masks indicating the locations of anomalies and enhance the network's ability to discriminate and localize defects by incorporating a segmentation head. However, it often suffers from overfitting problems, where the synthetic samples may be far from the real defects. A simple classification network can achieve nearly 100% accuracy in distinguishing between synthesized anomaly samples and real anomaly samples after only a few training epochs. Training with such a set results in an unclear discriminator that generalizes poorly to real anomalies, where indistinct objects could be included in inlier feature space, causing large false-positive (FP) detections, as shown in Fig. 1(b). The discriminative model shows improvement but still struggles to accurately reconstruct anomalous not covered by the simulation.

To simultaneously address the above-mentioned problems, we propose a new feature-level denoising autoencoder explicitly for surface AD, dubbed as **FeatDAE**, as illustrated in Fig. 2. Our FeatDAE introduces feature radial transformation (FRT) and feature omnidirectional injection (FOI) in the latent space after the encoder and incorporates a meticulously symmetrical decoder to seamlessly restore the latent noised representation back to the reconstruction result. By concurrently introducing distinct perturbations at both the image and feature levels, our FeatDAE is proficient in discerning the intrinsic information and distribution of the data, showcasing enhanced robustness when addressing a myriad of real-world anomalies. Second, the objects we reconstruct

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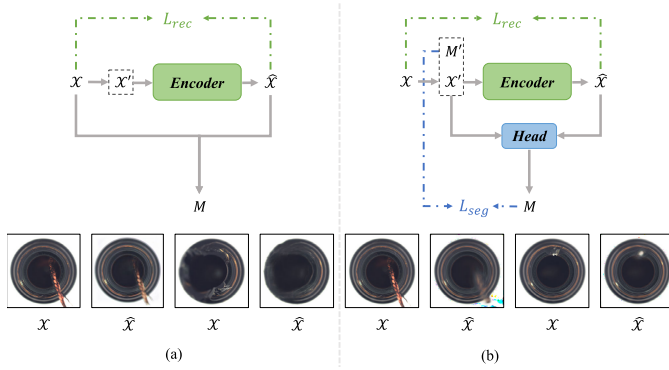


Fig. 1. Pilot study on reconstruction-based methods. (a) Generative model takes the original image $\mathcal{X}$ or the noise-added image $\mathcal{X}'$ as input, and is trained using the reconstruction loss L_{rec} to reconstruct its input. The anomaly map $\mathcal{M}$ is obtained by comparing the input $\mathcal{X}$ with the reconstruction result $\hat{\mathcal{X}}$. (b) Discriminative model combines the synthesized image $\mathcal{X}'$ with the reconstruction result $\hat{\mathcal{X}}$, and is supervised by the corresponding synthesized mask $\mathcal{M}'$ and the segmentation loss L_{seg} to produce a more accurate localization of defects. As shown, the minor discrepancy between the input $\mathcal{X}$ and the reconstructed result $\hat{\mathcal{X}}$ renders it difficult to accurately discern and differentiate anomalies. This highlights the challenge inherent in reconstruction-based methods, which stems from their limited capacity to distinctly characterize real anomaly samples.

encompass information of various granularity and types, such as pixel color, hand-crafted descriptors, and deep features. The diversity in the reconstructed targets further contributes to a comprehensive and precise depiction of features, thereby enhancing the downstream AD performance. In addition, we have devised a feature alignment module to address the varied sizes and morphologies of the reconstructed features. The aligned features are then fed into the head to handle discrepancies in a trainable manner, alleviating the overfitting of synthetic defects.

The main contributions of our paper are summarized as follows.

- 1) We introduce stochastic perturbations in both the input and latent space, extending and generalizing the concept of the denoising autoencoder. This approach directly addresses the issue of overfitting to simulated defects, a common problem in recent discriminative methods.
- 2) We propose a multifeature reconstruction mechanism that enables the autoencoder to learn from different types of features. The aligned features demonstrably mitigate the issue of overgeneralization and reduce the high FN rate.
- 3) We introduce a novel framework, FeatDAE, specifically designed for surface AD. FeatDAE is capable of handling a wide variety of anomalies effectively, making it versatile and adaptable to different applications.
- 4) Extensive experiments on the challenging MVTec AD dataset [12] and the VisA dataset [13] demonstrate the effectiveness and versatility of our method across various scenarios. Our FeatDAE sets a new state of the art in AD, achieving 96.3% region-wise AUPRO and 81.5% pixel-wise AP on the MVTec AD dataset, and 93.3% region-wise AUPRO and 47.1% pixel-wise AP on the VisA dataset. These results surpass the previous

state-of-the-art methods by a significant margin while maintaining a high inference speed of 110 FPS.

II. RELATED WORKS

AD: has been the focus of extensive research in recent years, and a plethora of approaches have been proposed to tackle this challenging task in industry. These methods can be broadly categorized into two main paradigms: reconstruction-based and embedding-based.

Reconstruction-based methods involve training an autoencoder network [6], [9], [14], [15], [16], [17] or a generative model [4], [18], [19] and operate under the assumption that anomalies will be poorly reconstructed, thereby making them distinguishable by their reconstruction error. However, due to the excellent generalization capabilities of the autoencoder, this assumption does not always hold, causing suboptimal performance. Some recent works [6], [20], [21] involved masking parts of the image or adding random noise, leading to better robustness of the autoencoder. Some methods [1], [10], [11], [22], [23], [24] are inspired by supervised segmentation tasks, by synthetically generating defects during the training phase with the idea that the model can then generalize on real anomalies. However, this discriminative model often leads to overfitting the training set and does not generalize well to real defects.

Embedding-based methods utilize feature maps [25], [26], [27] extracted through a pretrained network to learn normality on these maps. Certain methods [28], [29], [30] operate under the premise that the characteristics of normal data conform to a multivariate Gaussian distribution. Consequently, these approaches leverage the normal training examples to estimate the pertinent parameters and assume they will not be effective for anomaly data. Moreover, a variety of approaches [31], [32], [33], [34] based on normalizing flow have emerged, which endeavor to approximate the density by projecting an arbitrary distribution onto a Gaussian distribution. Nevertheless, all these methodologies presuppose that the distribution of normal regions will be adequately represented within the training data, rendering them ineffective when confronted with rare normal regions that remain unseen during the training phase, thereby resulting in FPs while facing real anomalous samples.

Denoising autoencoder [35] is an extension of the basic autoencoder and is trained to use a corrupted version of the input to reconstruct the original input. A variety of methods [36], [37], [38], [39], [40] can be considered as a generalized denoising autoencoder. Building on advancements in natural language processing [41], [42], [43], recent studies [44], [45], [46] have shown that modern vision Transformers hold significant potential when applied to representation learning. The inherent disadvantage of two-stage methods stems from their reliance on a pretrained VAE [47], tasked with transforming originally continuous data into intentionally discretized target visual tokens. Conversely, end-to-end training for masked autoencoders [48], [49] significantly enhances the model's performance in terms of both efficacy and efficiency. Our approach further adds noise at the feature level to improve the generalization of the model while maintaining accuracy and efficiency.

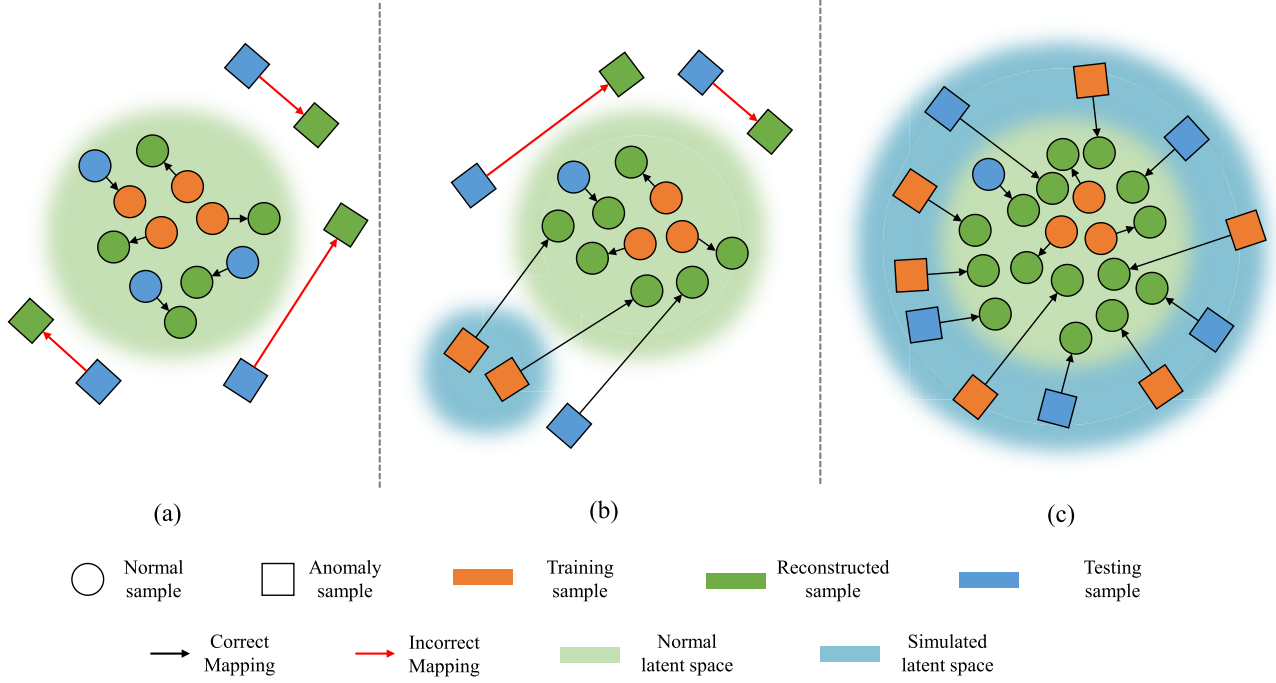


Fig. 2. Comparison of different architectures for defect detection and localization. (a) Generative model exhibits overgeneralization, leads to a failure in accurately localizing regions of test-time anomaly samples. (b) Discriminative model struggles to reconstruct real-world anomalies that differ significantly from the simulated ones, highlighting its limitations in handling unseen or more complex anomalies. (c) Our proposed FeatDAE addresses these issues by applying transformations at the feature level, significantly expanding the simulated space of anomalous and mitigating the overfitting problem. By reconstructing multiple types of features, our method achieves a more robust and accurate AD.

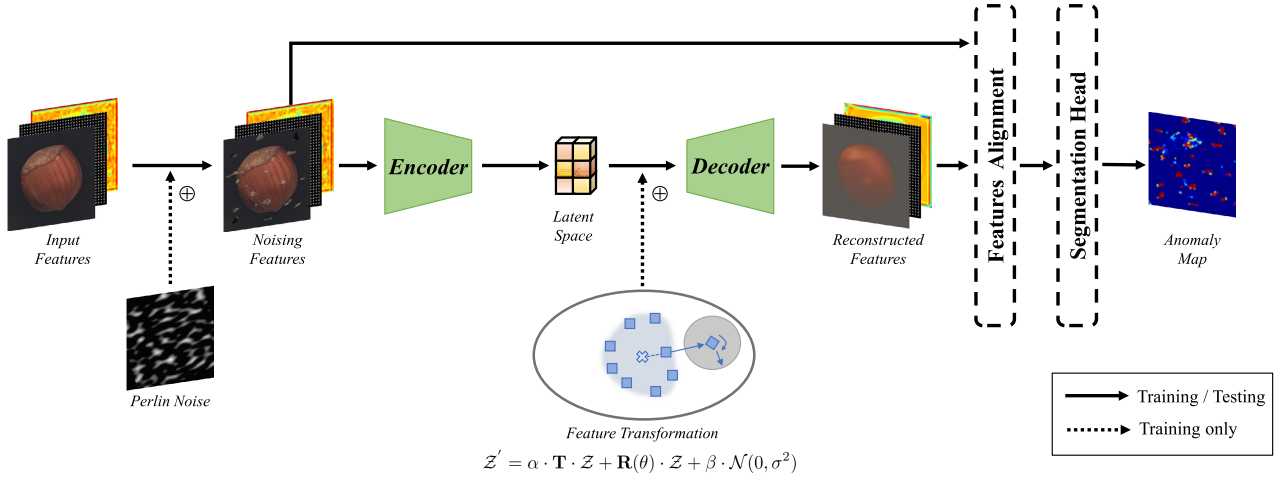


Fig. 3. Overview of our framework. By adding Perlin noise at the input space and making feature transformation at the latent space, our FeatDAE is able to learn more essential features of the sample. The reconstructed features are sent to the feature alignment module along with the noising features, and the final result is obtained after the segmentation head.

III. METHODOLOGY

A. Framework

Let $\mathcal{I}^{\text{train}} = \{I_1^{\text{train}}, \dots, I_n^{\text{train}}\}$ be the training set with anomaly-free samples only, and $\mathcal{I}^{\text{test}} = \{I_1^{\text{test}}, \dots, I_m^{\text{test}}\}$ be the testing set containing both normal and abnormal images. The goal of AD is to learn from the normal samples in the training set to accurately identify and locate faults when presented with a test set. Normal samples in both the training and test sets are considered to follow the same pattern. Any out-of-distribution is deemed abnormal. By juxtaposing input samples of unknown normalcy with previously modeled

normal samples, the derivation of a conclusive anomaly map is achieved.

Fig. 3 depicts the proposed FeatDAE framework for AD. At the training stage, the original input image $I^{\text{train}} \in \mathcal{I}^{\text{train}}$ is first fed into a freeze features extractor to obtain normal histogram of oriented gradient (HOG) feature $\mathcal{X}_{\text{HOG}}$ and deep features $\mathcal{X}_{\text{deep}}$. Second, the image-level noise is added to the input image and together they are passed into the encoder. Furthermore, our proposed FRT and FOI are integrated into the latent space to enrich and simulate a diverse array of defect characteristics. The decoder generates the corresponding multimodal features $\{\widehat{\mathcal{X}}_{\text{pix}}, \widehat{\mathcal{X}}_{\text{hog}}, \widehat{\mathcal{X}}_{\text{deep}}\}$, which are sent into the

feature alignment module together with the previous noising features $\{\mathcal{X}'_{\text{pix}}, \mathcal{X}'_{\text{hog}}, \mathcal{X}'_{\text{deep}}\}$. These aligned features are finally optimized in the segmentation head to localize the anomalous regions. For inference, the denoising autoencoder takes the input image $I^{\text{test}} \in \mathcal{I}^{\text{test}}$ to restore the features $\hat{\mathcal{X}}$. The pixel-level anomaly map is generated in an end-to-end manner with the aligned input features $\mathcal{X}$ and reconstructed features $\hat{\mathcal{X}}$, and the corresponding image-level anomaly scores are computed through postprocessing.

B. Feature-Level Denoising Autoencoder

Autoencoder is a neural network that is trained to reconstruct its input [50], [51]. The main purpose is to learn, in an unsupervised manner, an “informative” representation of the data that can be used for various implications such as clustering. Let $\mathcal{X}$ be the input data and $\hat{\mathcal{X}}$ be the output data, the aforesaid can be formulated as follows:

$$\hat{\mathcal{X}} = F(\mathcal{X}) \quad (1)$$

where F is the function of the encoder. As formally defined in [52], the distance between the reconstructed samples $\hat{\mathcal{X}}$ and the inputs $\mathcal{X}$ is regarded as the optimization objective. After being well-trained on anomaly-free sets, the expected outcome of the autoencoder is to accurately reconstruct normal samples but not abnormal ones. Therefore, the distance between each pixel in the input and output can be directly treated as an anomaly map.

Denoising autoencoder [35] can be viewed as a regularization option, which can extract more crucial information compared to stander autoencoder [53]. As the difference in the reconstruction errors in anomalous and nonanomalous regions increases, the downstream AD results improve [6]. Similar to an ordinary autoencoder, a denoising autoencoder can be represented as

$$\hat{\mathcal{X}} = F(\mathcal{X}') \quad (2)$$

where $\mathcal{X}'$ are generated from $\mathcal{X}$ by noise addition.

However, adding noise at the pixel level has several limitations. We observe that methods for simulating noise [1], [10] inevitably suffer from overfitting. Specifically, while they can effectively reconstruct noisy training images, they struggle to accurately reconstruct real anomaly images. In other words, the pixel-level noise, initially intended to enhance generalization, has paradoxically become a source of overfitting. Simultaneously, the images themselves contain inherent redundancies. For an extreme example, consider shifting all columns to the right by one pixel; while the ℓ_2 distance between the old and new images would significantly increase, the intrinsic information conveyed by the image remains unchanged. The encoder typically extracts features from the pixels to obtain a high-dimensional representation $\mathcal{Z}$, which simplifies and decouples the data to some extent. Hence, we argue that manipulating features in the latent space can be more effective in increasing the separation between normal and abnormal samples, thereby better accommodating the various defects encountered in real-world scenarios. For instance, we use ResNet [54] as the encoder and decoder architecture, and the model consists of several residual-like blocks to ensure

mirror symmetry. Specifically, the encoder downsamples the input image into latent space $\mathcal{Z}$ by four residual blocks while the corresponding decoder replaces all downsampling with bilinear upsampling restores the noise-added $\mathcal{Z}'$.

To mitigate the aforementioned issues, we have proposed **FeatDAE**, a feature-level denoising autoencoder. Inspired by the success of denoising autoencoders in enhancing network generalization by introducing noise in the pixel space, our method operates on features within the latent space. Specifically, **FeatDAE** applies two operations in the latent space: FRT and FOI. Our goal is to increase the diversity of samples in the latent space, thereby reducing overfitting to pixel-level synthetic defective samples and guiding the network to focus its generalization on real anomaly samples.

1) *Feature Radial Transformation*: For a batch containing B identities, where each feature map $\mathcal{Z} \in \mathbb{R}^{C \times H \times W}$, we notice that traditional convolution kernels often focus only on local information, which may not fully capture global context even after passing through multiple layers of deep networks. To enhance the model’s global understanding and increase diversity in the latent space, we introduce a radial movement to each feature, moving it away from the batch samples’ hidden space centroid. The centroid of a feature map can be defined as the average spatial coordinates of its nonzero elements. For simplicity, we assume the centroids are calculated over the spatial dimensions (H, W) for each channel c

$$\mathbf{c}_i = \left(\frac{\sum_{h=1}^H \sum_{w=1}^W h \cdot \mathcal{Z}_i(h, w)}{\sum_{h=1}^H \sum_{w=1}^W \mathcal{Z}_i(h, w)}, \frac{\sum_{h=1}^H \sum_{w=1}^W w \cdot \mathcal{Z}_i(h, w)}{\sum_{h=1}^H \sum_{w=1}^W \mathcal{Z}_i(h, w)} \right). \quad (3)$$

The center $\bar{\mathcal{Z}} \in \mathbb{R}^{C \times H \times W}$ for the batch samples is computed as the mean across all feature maps

$$\bar{\mathcal{Z}} = \frac{1}{B} \sum_{i=1}^B \mathcal{Z}_i \quad (4)$$

where i indexes the individual feature maps within the batch. Similarly, the centroid of the center $\bar{\mathcal{Z}}$ is

$$\bar{\mathbf{c}} = \left(\frac{\sum_{h=1}^H \sum_{w=1}^W h \cdot \bar{\mathcal{Z}}(h, w)}{\sum_{h=1}^H \sum_{w=1}^W \bar{\mathcal{Z}}(h, w)}, \frac{\sum_{h=1}^H \sum_{w=1}^W w \cdot \bar{\mathcal{Z}}(h, w)}{\sum_{h=1}^H \sum_{w=1}^W \bar{\mathcal{Z}}(h, w)} \right). \quad (5)$$

The translation vector (t_x, t_y) of the translation matrix $\mathbf{T}$ for the i th sample is calculated as the difference between the centroid of the feature map $\mathbf{c}_i$ and the centroid of the center $\bar{\mathbf{c}}$

$$(t_x, t_y) = \mathbf{c}_i - \bar{\mathbf{c}}. \quad (6)$$

This ensures that the translation moves the feature map away from the center in the direction from $\bar{\mathbf{c}}$ to $\mathbf{c}_i$, with the magnitude being a proportion α of the distance between the two centroids. The translation matrix $\mathbf{T}$ is addressed by

$$\mathbf{T} = \begin{bmatrix} 0 & 0 & t_x \\ 0 & 0 & t_y \\ 0 & 0 & 1 \end{bmatrix}. \quad (7)$$

Rotation can break this locality, acting as a form of “feature rearrangement” that changes the spatial structure of features and forces the model to understand and integrate these changes

at a higher level, which is especially important for latent space features requiring global understanding. Specifically, when we randomly select an angle θ to rotate the extracted 2-D feature map, the rotation transformation around the origin by an angle θ can be represented by the following matrix:

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (8)$$

To combine the rotation and translation into a single affine transformation, we multiply the rotation matrix $\mathbf{R}(\theta)$ by the translation matrix $\mathbf{T}$ with a scale factor $\alpha \in [0, 1]$. Applying the combined affine transformation to the original coordinates (x, y) in the feature map $\mathcal{Z}_i$ gives us the final transformed coordinates (x', y')

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & \alpha \cdot t_x \\ \sin(\theta) & \cos(\theta) & \alpha \cdot t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (9)$$

2) Feature Omnidirectional Injection: Our approach aims to expand the distribution of anomalies in the hidden space. Inspired by the widely adopted practice of adding noise at the image level for data augmentation [55], we introduce Gaussian noise with a mean of 0 directly into the latent space. As the depth of neural networks escalates, they are predisposed to the vanishing gradient problem, a phenomenon where the gradients diminish exponentially as they are backpropagated through the layers, which can impede the learning process and degrade the performance of deep models [54]. In the context of our proposed method, noise introduced at the input image level tends to have a diminishing impact as it is propagated through the network. By the time this noise reaches the latter stages of the network, particularly the decoder, its influence is substantially attenuated. This observation motivates us to introduce noise directly into the latent space, an approach that capitalizes on the inherent robustness of the deeper decoder layers. We aim to augment the model's ability to generalize from the training data to unseen data, especially when faced with real-world anomalies that were not present during training. The injection of noise into the latent space serves as a form of regularization, which not only helps mitigate the risk of overfitting but also improves the model's resilience against various types of anomalies. Consequently, this strategy enhances the overall robustness and adaptability of the computer vision system, making it more reliable and effective in practical defects.

Unlike the introduced FRT, which imposes a directional bias on the feature transformation, the FOI does not specify the transformation direction. It diffuses features uniformly across all dimensions, leading to a more isotropic distribution of the anomalies in the latent space. To control the intensity of the Gaussian noise, we employ a random scaling factor β , allowing us to modulate the magnitude of the perturbation. This is achieved according to the following formula:

$$\mathcal{Z}'' = \mathcal{Z}' + \beta \cdot \mathcal{N}(0, \sigma^2) \quad (10)$$

where $\mathcal{Z}'$ represents the latent representation after the previous FRT, $\mathcal{N}(0, \sigma^2)$ denotes the Gaussian noise, and β is a randomly determined coefficient that scales the strength of the

noise. By adjusting β , we can effectively balance the degree of perturbation and the preservation of the feature structure, ensuring that the data remains representative of the underlying distribution while providing sufficient variability to train a more generalized model.

C. Reconstruction Targets

For autoencoder based on reconstruction, a crucial issue lies in selecting the target feature $\mathcal{X}$ for reconstruction. We simultaneously consider reconstruction at different levels of granularity, including pixel features $\mathcal{X}_{\text{pix}}$ directly from the image samples, histogram of oriented gradients (HOG) features $\mathcal{X}_{\text{hog}}$ extracted through traditional operators, and deep features $\mathcal{X}_{\text{deep}}$ obtained through a pretrained neural network.

Pixel color is the most simple and straightforward target for feature reconstruction. In various similar tasks such as image inpainting [37] and mask prediction [56], pixel color reconstruction has been widely employed. Specifically, we treat the image from the dataset $\mathcal{I}$ as the reconstruction target $\mathcal{X}_{\text{pix}}$ for the autoencoder. Unlike the masked image modeling approaches [48], [49] that only count the masked area, our method computes the loss of all pixels.

HOG is a feature descriptor widely used in computer vision and image processing for object detection. It characterizes local object appearance and shape based on gradient orientations within an image. Given an image I , HOG first computes the gradient information

$$G_x(x, y) = p(x + 1, y) - p(x - 1, y) \quad (11)$$

$$G_y(x, y) = p(x, y + 1) - p(x, y - 1) \quad (12)$$

where $G_x(x, y)$ and $G_y(x, y)$ are the horizontal gradient and vertical gradient at pixel $p(x, y)$, respectively. The gradient magnitude $G(x, y)$ and orientation $\theta(x, y)$ are calculated as

$$\nabla G(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2} \quad (13)$$

$$\theta(x, y) = \arctan\left(\frac{G_y(x, y)}{G_x(x, y)}\right). \quad (14)$$

The next process involves dividing the image into small connected regions called cells and computing histograms for each cell. Specifically, the gradients within each cell are then accumulated into orientation histogram vectors of several bins, voted by gradient magnitudes. The final HOG descriptor is the concatenation of all normalized histograms from the cells, which captures local texture patterns and edge directions. Since HOG operates on localized cells of the image, it maintains good invariance to geometric and photometric transformations. The coarse spatial sampling and fine orientation sampling allow robustness while strong local normalization handles illumination changes. The efficiency and robustness to various conditions make HOG suitable for pattern recognition [57], [58] and feature reconstruction [59]. We collect HOG in each RGB channel to include color information which can improve its performance. The image samples $\mathcal{I}$ that computed the HOG feature map on the whole image are formed into $\mathcal{X}_{\text{hog}}$. Our method then learns to restore the histograms $\mathcal{X}_{\text{hog}}$ from the input noising image $\mathcal{X}'$.

Deep features are obtained by extracting intermediate layer features from a neural network. The neural network transforms

the input data into a more expressive form, which is capable of capturing the essence of the data across various contexts. Taking inspiration from knowledge distillation [60], we design our student networks to learn from the teacher networks similarly. However, unlike knowledge distillation, our method involves different inputs for the teacher and student networks. Our DAE serves as the student network and performs denoising reconstruction on input anomaly images, whether they are real anomalous or noise-added. In contrast, the teacher network processes anomaly-free images. Further, to better leverage the semantic information from low-resolution to high-resolution, we proposed a layer-by-layer deep features reconstruction hierarchy. In our method, we use an ImageNet [61] pretrained ResNet [54] as our feature extractor to derive features from normal images. Specifically, the deep features $\{\mathcal{X}_{\text{deep}}^2, \mathcal{X}_{\text{deep}}^3, \mathcal{X}_{\text{deep}}^4\}$ are obtained from the conv2_x, conv3_x, and conv4_x blocks with strides of $\{4, 8, 16\}$ pixels relative to the input image.

D. Features Alignment Module

As illustrated in Section III-C, our reconstruction targets incorporate a variety of distinct feature types. Pixel color features have the lowest dimensionality and primarily capture the original texture information necessary for reconstructing anomaly-free samples. In comparison, HOG features are more robust to variations in lighting and are capable of capturing edge and shape information within the image. While both pixel color and HOG features ensure detail preservation at larger scales, the deep features of different sizes predominantly contain different levels of semantic information, which is critical for subsequent segmentation tasks. Therefore, to handle the features discriminability varies across different levels of abstraction, we proposed the features alignment module to better integrate features between input $\mathcal{X}$ and reconstructed $\hat{\mathcal{X}}$ as described in Fig. 4. Each lateral connection fuses the input features and reconstructed features of the same spatial size and type form and computes the corresponding reconstruction loss L_{rec}^k . These merged features are then resized and concatenated into the aligned features used for subsequent segmentation heads.

Through element-wise multiplication, the paired features are merged without change in dimension. We also investigate alternative ways to compute the aligned features in Section IV-F. Among the multiple feature targets, pixel color and HOG have the same size as the original image, while deep features include multiple scales. We uniformly scaled all the features into 1/4th of the input size, the same size as the largest feature map in deep features. While this scale results in loss of detail to some extent, oversizing exponentially increases the amount of computation and memory usage, which is critical for real-time applications.

E. Objective

As mentioned earlier, our feature space varies in type and dimensions. We follow [7] of using cosine distance for knowledge transferring to more accurately capture the relationship between high- and low-dimensional information [65]. At each

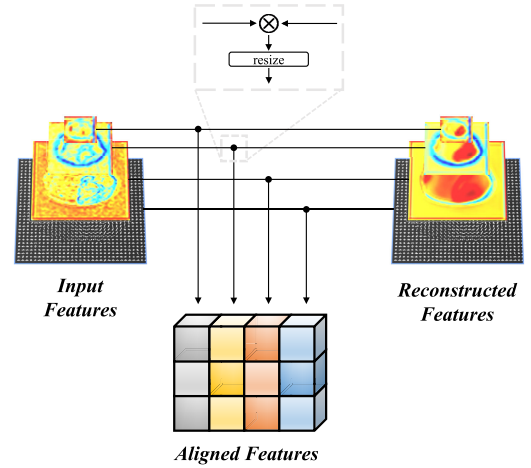


Fig. 4. Features alignment module. At each scale, the input and reconstructed features are fused through pixel-wise multiplication. The merged features then undergo a resizing operation to ensure uniformity in size, followed by concatenation along the feature dimensions. The spectrum of multiscale features comprises pixel color, HOG, and deep features.

scale k , we calculate their vector-wise cosine similarity loss along the channel axis

$$L_k^{\cos} = 1 - \frac{\mathcal{X}^k \cdot \hat{\mathcal{X}}^k}{\max(\|\mathcal{X}^k\|_2, \epsilon) \cdot \max(\|\hat{\mathcal{X}}^k\|_2, \epsilon)} \quad (15)$$

$$L_{\text{rec}}^{\cos} = \sum_{k=1}^K L_k^{\cos} \quad (16)$$

where ϵ is a small value to avoid division by zero and k is the target features number.

We follow DRAEM [1] training a segmentation head guided by additional supervision to discriminate the fusion features appropriately. By concurrently inputting the reconstructed results and the original image into the head, employing the mask as the supervise for training the anomaly map as follows:

$$L_{\text{seg}}^{\text{dis}} = \ell_1(\mathcal{M}, \hat{\mathcal{M}}) \quad (17)$$

where $\mathcal{M}$ and $\hat{\mathcal{M}}$ are the ground-truth binary masks generated from noise and the output anomaly segmentation masks from the head, respectively. Since defects usually make up only a small portion of the image, only a minority of pixels are considered foreground for the segmentation task. Focal loss [66] allows the model to focus more on hard samples and helps to solve the problem of background and foreground imbalance

$$L_{\text{seg}}^{\text{focal}} = -(1 - p)^\gamma \log(p) \quad (18)$$

$$p = \mathcal{M}\hat{\mathcal{M}} + (1 - \mathcal{M})(1 - \hat{\mathcal{M}}) \quad (19)$$

where γ is the focusing parameter.

Along with the reconstruction loss and the segmentation losses mentioned above, the total objective of our network is formulated as follows:

$$L = L_{\text{rec}}^{\cos} + L_{\text{seg}}^{\text{dis}} + L_{\text{seg}}^{\text{focal}}. \quad (20)$$

Specifically, we normalize each loss term by dividing it by its maximum value across the training dataset. This normalization ensures that no single loss term dominates the others, allowing for a more stable and effective training process.

TABLE I
COMPARISON IN IMAGE-WISE AUROC AND PIXEL-WISE AUROC WITH STATE-OF-THE-ART WORKS ON THE MVTEC AD DATASET (I-AUROC/P-AUROC). WE DENOTE THE BEST RESULTS BY BOLD

Method	[1]	[7]	[32]	[62]	[10]	[63]	[24]	[64]	[8]	Ours
bottle	96.9/ 99.3	99.6/97.8	99.9/97.3	98.6/98.7	100.0 /95.7	99.8/98.2	98.5/98.1	100.0/99.4	97.8/93.5	100.0/99.3
cable	93.4/95.4	81.9/84.9	90.3/89.9	100.0/99.4	93.0/92.2	94.5/93.7	99.9/98.5	100.0 /99.4	85.9/90.8	98.4/ 98.0
capsule	96.1/94.0	98.2/98.8	87.8/98.5	98.3/98.5	91.3/86.0	97.6/98.8	100.0 /99.0	99.5/99.7	97.9/90.3	98.2/ 99.1
carpet	96.2/96.2	98.4/99.0	99.2/98.9	100.0/98.1	93.6/96.1	97.4/98.8	100.0 /94.9	100.0/98.4	91.3/87.7	99.8/98.1
grid	100.0/99.5	97.8/99.2	89.4/93.6	95.6/93.0	99.8/99.3	99.1/99.1	99.4/92.6	98.9/97.7	99.7/85.7	100.0/99.5
hazelnut	100.0 /99.5	100.0 /98.6	99.9/98.5	96.5/92.6	99.7/97.9	93.4/98.7	100.0 /98.2	100.0/99.1	96.6/95.3	100.0/99.7
leather	100.0 /98.9	100.0 /99.3	100.0 /99.3	99.5/98.2	100.0 /99.5	100.0 /99.2	99.9/98.1	99.7/98.4	97.4/91.5	100.0/99.8
metal_nut	99.4/98.7	59.9/92.1	97.8/96.0	99.2/95.8	100.0 /96.9	100.0 /96.6	99.4/98.9	98.1/98.7	94.6/94.4	98.0/ 99.2
pill	96.7/97.6	97.0/97.5	85.0/96.5	99.8/98.3	88.9/89.5	97.6/98.2	99.7/98.0	99.8 /99.2	94.7/93.0	98.6/ 99.3
screw	98.7/ 99.5	98.3/99.4	71.4/97.0	99.5/99.0	86.4/74.4	98.7/99.5	100.0 /98.8	100.0 /98.8	72.4/86.4	96.9/99.4
tile	100.0/99.5	97.9/95.2	99.7/95.9	97.1/99.0	100.0 /98.9	99.9/96.3	98.0/98.4	96.6/97.6	93.0/76.2	100.0 /99.2
toothbrush	99.7/98.1	35.8/97.3	83.3/98.2	96.5/99.3	93.9/97.5	96.7/99.1	96.2/99.0	94.9/99.3	95.0/89.7	100.0/99.5
transistor	94.2/90.0	93.7/85.6	92.4/84.6	100.0/98.7	99.5/76.6	95.4/89.5	91.1/98.4	100.0 /99.2	99.0/98.1	99.8/ 98.9
wood	99.1/97.2	99.4/95.5	98.9/94.4	99.8/95.4	99.2/94.1	99.3/95.3	99.7/97.2	99.4/98.1	93.5/83.1	99.9/98.5
zipper	99.7/98.6	99.5/98.4	96.0/98.0	99.8/98.0	99.4/94.7	68.2/98.3	99.8/98.4	98.1/97.7	71.8/84.0	100.0/99.3
Average	98.0/97.5	90.5/95.9	92.7/95.8	98.7/97.5	96.3/92.6	95.8/97.3	98.8/97.8	99.0/98.7	92.0/89.3	99.4/99.1

TABLE II
COMPARISON IN PIXEL-WISE AP AND REGION-WISE AUPRO WITH STATE-OF-THE-ART WORKS ON THE MVTEC AD DATASET (AP/AUPRO). WE DENOTE THE BEST RESULTS BY BOLD

Method	[1]	[7]	[32]	[62]	[10]	[63]	[24]	[64]	[8]	Ours
bottle	89.7/85.3	68.0/94.6	62.3/92.1	74.0/94.1	88.0/92.6	71.0/94.8	42.2/91.4	78.1/97.0	45.9/75.9	90.9/97.1
cable	62.4/86.7	25.9/76.8	27.7/79.4	42.8/97.0	62.8/70.9	37.2/86.4	32.4/93.6	39.9/97.6	27.2/64.4	72.2/90.8
capsule	42.7/91.8	44.2/96.0	39.9/93.3	36.0/96.0	57.5/59.0	45.4/95.7	36.5/96.6	60.0/97.2	16.2/54.2	59.2/95.6
carpet	63.8/85.6	57.7/95.6	59.0/94.7	80.6/91.8	72.3/94.2	52.1/94.5	59.7/80.9	75.6/96.7	16.9/65.5	77.4/96.8
grid	55.3/91.3	45.9/97.3	21.3/82.3	61.9/79.2	65.9/97.8	43.7/96.7	40.3/80.5	70.9/95.0	3.9/57.4	71.6/98.4
hazelnut	87.7/89.7	58.3/96.4	59.9/95.7	52.0/76.3	90.2/90.5	57.1/96.4	67.9/91.3	86.0/97.3	27.9/81.0	94.6/98.0
leather	69.0/87.2	38.2/97.8	48.8/98.3	71.1/89.9	80.6 /98.8	39.7/97.4	66.4/91.0	68.2/93.5	6.4/77.9	78.4/99.2
metal_nut	91.4/88.7	50.0/87.5	71.4/88.8	34.5/89.4	91.2/93.2	74.0/93.4	41.6/93.1	45.4/96.1	73.7/53.9	94.8/94.4
pill	46.0/88.4	63.1/96.0	56.2/90.9	51.1/87.1	83.5/48.9	73.4/96.3	45.0/93.9	68.3/91.1	50.4/65.7	86.5/97.1
screw	70.6 /90.5	41.3/97.1	17.2/89.3	92.8/91.1	49.2/50.1	41.7/97.7	89.2/90.8	89.7/96.3	3.1/55.4	65.8/ 96.0
tile	96.9 /88.4	48.4/85.7	55.9/86.7	84.0/93.1	95.9/ 98.0	52.8/88.7	79.5/92.4	67.8/96.7	25.0/50.5	95.4/97.9
toothbrush	53.4/88.1	21.3/86.1	46.0/85.7	38.1/ 96.6	74.5/81.1	57.0/92.2	30.5/94.8	26.2/95.5	11.1/64.9	77.4 /95.1
transistor	51.3/87.8	41.7/74.8	38.8/74.1	43.3/ 93.5	60.4/78.2	48.8/78.3	50.1/82.4	46.5/96.1	72.8/85.3	84.9 /92.5
wood	80.8/91.5	49.4/91.6	47.0/91.0	67.0/90.8	83.6/94.8	50.2/90.3	65.9/92.3	78.9/92.0	15.0/58.0	87.1 /97.9
zipper	72.0/88.1	52.4/95.2	50.1/92.8	57.2/92.8	80.8/90.9	50.8/94.5	50.5/94.2	54.9/90.5	13.5/55.5	86.7 /97.3
Average	68.7/88.6	47.1/91.2	46.8/89.0	59.1/90.6	75.8/82.6	53.0/92.9	53.2/90.6	63.8/95.2	27.3/64.4	81.5 /96.3

IV. EXPERIMENTS

A. Dataset

We conducted our experiments on the most recent challenging MVTEC AD dataset [12] and the popular VisA dataset [13], as the standard benchmarks for evaluating unsupervised surface AD and localization. The MVTEC AD dataset consisted of five texture and ten object categories stemming from manufacturing, with 3629 training images and 1725 test images. The image size ranged from 700×700 to 1024×1024 pixels. The training split was composed of normal images, and the testing split contained both normal and anomaly images with various types of defects. It also provided detailed pixel-wise mask annotations for defective test images. The VisA AD dataset [13] was a challenging collection of images designed for evaluating machine-learning models' performance in identifying anomalies. It consisted of 10 821 images in total, with 9621 normal images and 1200 abnormal images. The dataset was organized into 12 subsets, each representing a distinct class of objects. Abnormal images within the MVTEC and

VisA datasets exhibited a range of defects, including surface imperfections like scratches, dents, colored spots, or cracks, as well as structural issues such as misplacement or missing components. The dataset was also divided into normal images in the training set and labeled normal and abnormal images in the testing set, making it suitable for unsupervised AD approaches.

B. Evaluation Metrics

In prior works, the area under the receiver operating curve (AUROC) was utilized to evaluate image-wise AD and pixel-wise anomaly localization. While only a minority of pixels were attributed to abnormal, the AUROC did not reveal the accuracy of localization under heavily imbalanced conditions [67]. For pixel-wise evaluation, we additionally reported the average precision (AP) [68] and area under per-region-overlap (AUPRO) [69], which expressed the performance for anomaly localization more effectively. The AUPRO assigned equal weight to connected components of different sizes in

TABLE III
COMPARISON IN IMAGE-WISE AUROC AND PIXEL-WISE AUROC WITH STATE-OF-THE-ART WORKS ON THE VISA DATASET (I-AUROC/P-AUROC). WE DENOTE THE BEST RESULTS BY BOLD

Method	[1]	[7]	[32]	[62]	[10]	[63]	[24]	[64]	[8]	Ours
candle	93.2/95.5	91.8/99.0	92.4/98.8	99.7/97.9	89.8/74.5	92.7/99.3	95.6/97.9	98.9/93.6	99.3/80.1	93.1/98.6
capsules	74.6/98.6	81.0/99.3	57.5/94.5	100.0/99.7	86.5/95.7	79.5/99.1	87.7/99.0	99.1/ 99.8	66.7/94.3	90.0/99.1
cashew	96.4/79.1	93.3/92.3	95.6/99.0	94.6/95.6	86.6/84.4	93.5/95.9	94.1/99.1	98.2/ 97.9	99.8/89.1	95.2/95.8
chewinggum	94.9/97.7	95.5/98.7	97.2/99.2	98.5/97.1	96.8/97.5	96.2/98.8	98.9/98.2	98.8/99.4	97.6/68.5	98.8/99.3
fryum	93.4/90.4	95.8/97.0	91.2/ 97.6	100.0/95.1	92.3/67.0	95.7/97.2	96.5/95.1	99.0/96.5	97.0/97.5	95.9/95.8
macaroni1	95.7/48.7	95.2/99.8	81.8/99.1	97.6/98.4	93.5/73.3	92.8/99.7	92.3/99.5	99.7/99.8	96.3/55.2	88.6/99.7
macaroni2	37.8/83.9	85.6/99.6	67.1/97.4	89.9/98.7	72.2/62.8	82.8/99.5	78.8/96.3	100.0/99.8	56.0/56.8	85.8/99.4
pcb1	65.7/96.5	96.4/99.5	94.9/99.1	100.0/91.0	78.6/96.0	96.3/ 99.4	98.7/99.6	99.7/99.2	91.7/95.7	97.0/98.9
pcb2	89.2/81.5	97.3/97.7	92.1/96.5	99.9/97.4	82.8/94.4	97.9/98.0	98.8/98.6	99.9/97.9	89.9/82.2	99.3/ 98.6
pcb3	94.1/98.3	96.8/98.0	80.8/96.4	100.0/97.2	95.8/90.6	95.7/98.0	98.0/98.8	100.0/98.5	94.0/87.2	97.0/ 99.3
pcb4	98.4/96.0	99.9/97.7	98.9/96.8	99.9/96.8	98.0/94.4	99.9/97.1	99.0/98.1	99.9/98.3	99.8/95.4	99.1/ 98.9
pipe_fryum	98.9/90.4	98.2/99.1	97.6/99.2	97.6/96.2	95.0/86.8	94.1/98.9	98.8/99.2	98.5/99.2	97.7/98.6	94.2/ 99.2
Average	86.0/92.3	93.9/98.1	87.2/97.8	98.1/96.8	89.0/84.8	93.1/98.4	94.8/98.3	99.3/98.3	90.5/83.4	94.5/98.6

TABLE IV
COMPARISON IN PIXEL-WISE AP AND REGION-WISE AUPRO WITH STATE-OF-THE-ART WORKS ON THE VISA DATASET (AP/AUPRO). WE DENOTE THE BEST RESULTS BY BOLD

Method	[1]	[7]	[32]	[62]	[10]	[63]	[24]	[64]	[8]	Ours
candle	33.5/93.1	19.1/95.2	14.7/93.7	17.1/95.7	39.5/49.9	25.5/95.4	7.6/80.4	22.7/86.8	1.0/41.4	42.3/97.1
capsules	28.8/85.7	57.3/91.9	27.9/65.9	67.4/97.9	38.7/77.8	56.8/91.4	21.2/57.5	49.4/96.3	8.0/55.3	38.2/ 95.4
cashew	26.0/78.0	44.9/84.5	52.5/94.6	37.6/71.1	48.2/55.8	56.3/85.1	68.0/85.0	43.3/93.0	45.6/21.5	85.5/92.3
chewinggum	34.8/76.6	58.4/78.1	61.9/87.8	27.5/88.0	78.3/68.4	58.8/80.1	39.7/74.7	65.7/94.7	15.6/18.6	46.7/ 87.0
fryum	44.5/79.4	48.1/93.1	53.8/87.4	39.5/90.7	47.6/35.4	49.9/92.0	40.6/78.5	34.0/93.4	59.0/68.9	46.1/89.4
macaroni1	48.7/81.8	23.6/97.2	7.9/95.2	8.3/98.0	15.2/36.6	24.0/96.7	1.5/66.5	17.1/98.3	0.0/3.6	23.3/ 97.3
macaroni2	2.3/78.8	12.3/97.0	1.9/89.9	16.0/98.6	10.3/36.0	9.6/96.2	0.4/66.4	20.4/98.6	0.0/15.8	10.3/97.8
pcb1	52.7/75.1	66.8/96.1	73.0/88.3	12.5/86.1	66.5/68.6	64.9/95.5	73.1/79.0	48.8/91.8	27.7/56.5	77.9/91.9
pcb2	0.8/67.6	21.1/91.7	12.6/82.5	13.9/93.1	19.1/69.8	25.5/91.7	7.2/76.0	9.0/90.7	4.4/48.1	29.5/92.3
pcb3	11.1/66.9	23.8/93.6	20.6/81.1	7.9/94.3	27.5/61.2	30.9/93.3	12.9/82.7	23.0/95.5	4.0/41.8	38.9/88.6
pcb4	27.3/65.9	30.0/88.4	22.2/85.6	43.2/87.4	53.8/74.9	27.4/85.8	13.9/67.1	38.8/92.9	9.9/82.4	39.9/ 88.5
pipe_fryum	21.5/65.0	55.1/95.9	61.6/94.9	43.8/81.1	74.3/55.9	54.9/94.2	64.1/76.6	56.9/97.3	55.1/77.9	90.3/96.2
Average	27.7/76.2	38.4/91.9	34.2/87.3	27.9/90.2	43.3/57.5	40.4/91.4	29.2/74.2	35.7/ 94.1	19.2/44.3	47.1/93.3

the ground truth (GT) and repeatedly computed the overlap between the prediction and the GT for the FP rate within 30%.

C. Implementation Details

Our implementation was based on the PyTorch framework. We used ResNet-18 [54] as our encoder and a symmetric reverse ResNet-18 architecture with upsampling instead of downsampling in all residual blocks as our decoder. Note that the encoder was pretrained on the ImageNet [61] dataset, while the decoder utilized the Kaiming initialization [70]. For the segmentation head, we adopted the widely used ASPP [71] structure. The learning rate was set to 0.001 on the SGD optimizer with a momentum of 0.9 and a weight decay of 0.001. We trained the denoising autoencoder for 4000 iterations and the segmentation head for 16000 iterations with a batch size of 8. We uniformly resized the images to 256×256 pixels, applied a random rotation of up to 5° for nonaligned targets (cable, wood, and zipper), and another random rotation within 360° for rotation-invariant textures (carpet, grid, leather, and tile) and objects (bottle, hazelnut, and screw). All images were normalized using the mean and standard deviation of ImageNet before feeding them into the model.

D. Quantitative Results

1) *Accuracy*: Table I quantitatively compared FeatDAE with recent state-of-the-art approaches on the MVTec AD

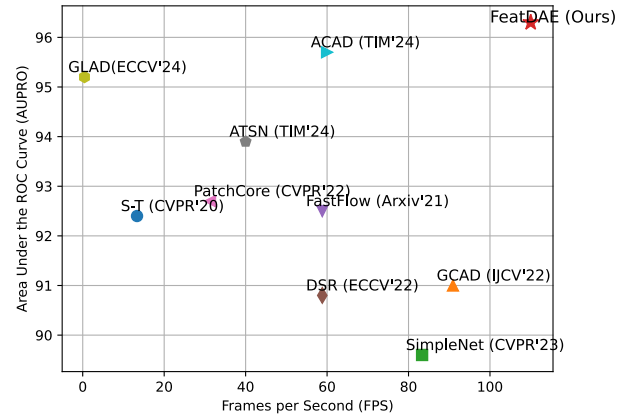


Fig. 5. AD performance versus FPS on the MVTec AD dataset, where our FeatDAE outperforms all previous methods on both accuracy and efficiency by a significant margin.

dataset under the AUROC metric. For the image-level AD task, FeatDAE achieved a perfect I-AUROC score in 7 out of 15 classes.

Table II further reported the pixel-wise AP and AUPRO for the pixel-level anomaly localization task, in addition to the P-AUROC metric. Overall, our FeatDAE also achieved

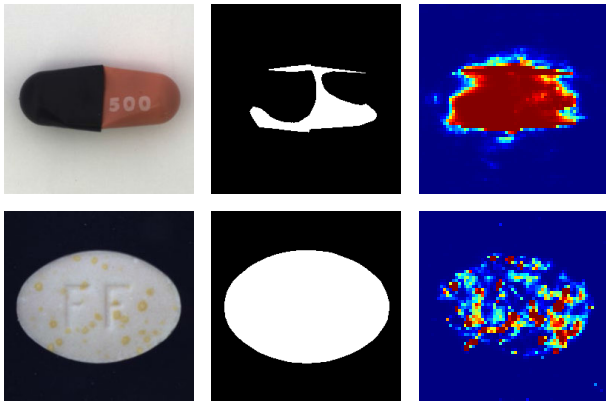


Fig. 6. Case of ambiguous anomalies on the MVtec AD dataset. From left to right: sampled image, GT, and anomaly map.

the best average results across these metrics. Our FeatDAE significantly outperformed all previous best-performing AD methods by a large margin. On average, our method surpassed the cutting-edge methods by 5.7% points in terms of pixel-wise AP. Our method achieved optimal performance in the majority of categories, demonstrating its applicability and robustness across various scenarios.

We also conducted experiments on the popular VisA dataset, which is a comprehensive benchmark for AD. We used the same metrics as the MVtec AD dataset to evaluate the performance across image-wise AUROC, pixel-wise AUROC, pixel-wise AP, and region-wise AUPRO. These metrics provided a holistic view of the model’s ability to detect anomalies at different levels of granularity. Notably, our FeatDAE utilized ResNet-18 [54] as the backbone, while previous methods [7], [24], [25] employed WideResNet-50 [72]. Therefore, our method could be more readily deployed on edge devices with suboptimal performance in actual industrial pipelines.

As can be seen in Table III, the image-wise AUROC and the pixel-wise AUROC were strong, indicating high accuracy in distinguishing between normal and anomalous images with the lightweight ResNet-18 backbone. Table IV reports the pixel-wise AP and AUPRO on the VisA dataset. In pixel-wise AP, FeatDAE also performed well, achieving an average score of 47.1%, ranking it among the top methods. Notably, our method ranked second in terms of region-wise AUPRO, with an average score of 93.3%. While the GLAD method [64] achieves high accuracy, it suffers from inefficiency due to the step-by-step generation process of diffusion models. The superior performance in these metrics underscored the exceptional capability of our method in accurately pinpointing anomalous. Our FeatDAE precisely localized the regions of interest, making it a powerful tool for applications requiring fine-grained AD.

2) *Efficiency*: Anomaly detector is commonly employed in industrial production lines, where the system must be lightweight to meet real-time processing demands. As illustrated in Fig. 5, we evaluate the model’s capability in accurately localizing defects using the AUPRO metric, and its operational efficiency is assessed by the frames per second (FPS). The figure demonstrates that our proposed method attains a balance between precision and speed, achieving optimal performance on both metrics.

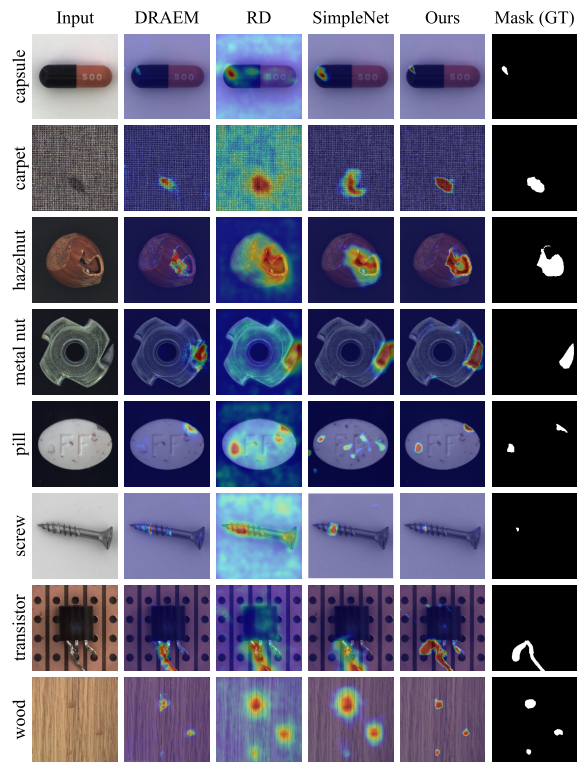


Fig. 7. Qualitative comparison on the MVtec AD dataset. Input image, anomaly map, and GT are shown.

3) *Ambiguity*: Upon meticulous examination, it was discerned that a portion of the detection inaccuracies could be ascribed to imprecise GT labels in cases of equivocal anomalies. This phenomenon was exemplified in Fig. 6, wherein the GT contained only the deformed sections of the capsule, yet it comprehensively included every aspect of the pill. Our method delineated the entire deformed area of the capsule and accurately segmented the yellow spots on the pill. However, discrepancies with the actual GT led to cumulative errors in the localization metrics. Despite this, we believed our approach remained viable in such scenarios.

E. Qualitative Results

We also compared the qualitative results of different models on the MVtec AD dataset, as illustrated in Fig. 7. Test set images are displayed to demonstrate each model’s performance on previously unseen anomalous samples. Accompanying these images are prediction anomaly maps, where red signifies high detection confidence and blue indicates low confidence. GT masks delineate the actual anomaly regions, enabling a direct comparison between predicted and true defect locations.

From the results, it is evident that our method achieves precise localization of all anomalies compared to alternative approaches. For smaller anomaly regions, such as those in the capsule and screw categories, RD [7] tends to overestimate the size of the mask region, while DRAEM [1] introduces FPs. In contrast, our method produces sharp and accurate boundaries. In cases of irregularly shaped defects, like those found in the hazelnut and transistor categories, other methods either underestimate the defect area or exhibit other inaccuracies. Our method, however, aligns closely with the true

TABLE V

ABLATION STUDIES ON OUR MAIN DESIGNS. DETECTION (DET.) RESULTS ARE REPORTED IN AUROC AND LOCALIZATION (LOC.) RESULTS ARE REPORTED IN AP. THE DEFAULT ENTRY IS MARKED AS GRAY

Exp.	Feature FRT	Trans. FOI	Recon. Pixel	Target HOG	Deep	Det.	Loc.
1	-	-	✓	✓	✓	98.5	74.0
2	-	✓	✓	✓	✓	99.3	80.9
3	✓	-	✓	✓	✓	99.2	78.2
4	✓	✓	-	✓	✓	99.2	79.9
5	✓	✓	✓	-	✓	99.3	78.8
6	✓	✓	✓	✓	-	98.7	73.6
7	✓	✓	✓	✓	✓	99.4	81.5

TABLE VI

ABLATION STUDIES ON FEATURES ALIGNMENT MODULE IMPLEMENTATION. THE DEFAULT ENTRY IS MARKED AS GRAY

type	add.	sub.	mul.	concat.
Det.	98.0	98.3	99.4	99.1
Loc.	73.8	75.9	81.5	79.6

FRT is absent but feature-level omnidirectional injection (FOI) is applied (Experiment 2), the model achieves a high detection score of 99.3% and a localization score of 80.9%, indicating that FOI alone significantly enhances the model’s robustness. Conversely, Experiment 3, where FRT is present without FOI, shows slightly lower performance metrics, suggesting that while beneficial, its impact does not match that of FOI in this context. Further insights are gained from Experiments 4 to 6, which omit pixel, HOG, and deep feature reconstructions, respectively. The importance of these features for maintaining high localization accuracy is evident, with the absence of HOG features (Experiment 4) leading to a decrease in localization performance from the optimal baseline of 81.5% down to 79.9% and excluding deep features (Experiment 6) resulting in a detection score of 98.7% and a localization score of 73.6%. This underscores the essential roles of HOG features in capturing texture and edge information and deep features in conveying higher level semantic details critical for precise anomaly localization. The default setup (Experiment 7), incorporating all design elements—FRT, FOI, and comprehensive feature reconstruction—achieves the highest detection AUROC of 99.4% and localization AP of 81.5%. This highlights the synergistic benefits of integrating all proposed strategies, optimizing both detection and localization capabilities.

2) *Operator in Feature Alignment Module*: Table VI presents the results of experiments conducted using different fusion operations including addition, subtraction, multiplication, and concatenation on the input features and reconstructed features illustrated in Fig. 4. The subtraction operation exhibited superior performance compared to addition in localization, as it simulated the calculation of the reconstruction loss and introduced more prior knowledge. The concatenation operator achieved higher results due to retaining complete information from both the input features and the reconstruction features. However, these operators were all inferior to the multiplication operator, indicating its suitability for our task. Excessive prior

TABLE VII

ABLATION STUDIES ON THE TYPE OF THE NOISE IN INPUT SPACE. THE DEFAULT ENTRY IS MARKED AS GRAY

type	rectangle	MAE	Gaussian	Perlin
Det.	95.6	98.2	98.7	99.4
Loc.	70.3	73.0	75.4	81.5

TABLE VIII

ABLATION STUDIES ON FEATURE-LEVEL NOISE SCALE. THE DEFAULT ENTRY IS MARKED AS GRAY

σ	0.01	0.02	0.04	0.08	(0.01, 0.04)
Det.	99.2	99.3	99.2	99.1	99.4
Loc.	78.7	80.1	78.4	78.6	81.5

knowledge or overly comprehensive information might lead to suboptimal network performance.

3) *Input Space Noise Type*: Table VII presents an ablation study on the impact of different types of noise added to the input space on detection performance. Rectangle simulates anomaly regions by adding rectangular masks to the input images. Masks of MAE are generated by splitting the image into a grid of rectangular regions. For Gaussian noise and Perlin noise, a threshold of 0.5 is applied to the noise before generating the final mask.

The Perlin noise achieves the highest detection and localization accuracy, outperforming other noise types. This suggests that Perlin noise, which generates more complex and varied patterns, better mimics real-world anomalies, allowing the model to learn more realistic and diverse defect patterns.

4) *Latent Space Noise Scale*: Table VIII revealed the effect of σ on AD. The minuscule σ signified that the features after passing through the encoder had not been endowed with sufficient noise, resulting in the denoising autoencoder as a whole failing to capture the intrinsic feature distribution of the samples.

Consequently, its performance during testing was subpar. However, an excessively large σ altered the original distribution of the extracted features, leading to inconsistent learning within the network, thus causing ambiguous decision-making. As can be seen from Table VIII, with the gradual increase in σ , the final detection and segmentation results exhibited a trend of first rising and then declining. Hence, we opted for sampling σ within the range of (0.01, 0.04) and adding feature-level Gaussian noise in this manner, as experiments demonstrated it yielded optimal results.

5) *Loss*: In Table IX, we investigated the influence of different loss functions. Utilizing traditional ℓ_1 or ℓ_2 distance functions led to a significant decrease in results, mainly due to our FeatDAE aggregating multiple feature reconstruction results for the final segmentation. The cosine distance function was better equipped to handle high-dimensional features, as our experiments validated the soundness of this choice.

Table X demonstrated the impact of different losses on the segmentation task. Given that defects typically occupied only a small portion of the sample, we employed focal loss [66] to balance positive and negative samples. Although in the case of using only a single type of loss, ℓ_2 yielded relatively favorable

TABLE IX
ABLATION STUDIES ON THE RECONSTRUCTION LOSS.
THE DEFAULT ENTRY IS MARKED AS GRAY

type	ℓ_1	ℓ_2	cosine
Det.	98.7	98.2	99.4
Loc.	74.9	75.3	81.5

TABLE X
ABLATION STUDIES ON THE SEGMENTATION LOSS. THE
DEFAULT ENTRY IS MARKED AS GRAY

type	ℓ_1	ℓ_2	focal	$\ell_1 + \text{focal}$	$\ell_2 + \text{focal}$
Det.	97.9	99.1	97.0	99.4	99.2
Loc.	68.9	77.2	72.4	81.5	77.3

TABLE XI
COMPARISON IN PIXEL-WISE AP, F_1 -MAX AND ITS CORRESPONDING
PRECISION AND RECALL WITH STATE-OF-THE-ART WORKS ON THE
MVTec AD DATASET. WE DENOTE THE BEST RESULTS BY BOLD

Method	[6]	[1]	[7]	[24]	[63]	Ours
AP	48.2	68.4	47.1	24.0	53.0	81.5
F_1	37.4	66.7	57.3	55.3	58.1	71.8
Precision	32.5	65.9	50.5	47.8	51.6	71.3
Recall	51.0	69.6	67.9	68.2	67.8	72.7

TABLE XII
COMPARISON ON CLASSIFICATION RESULTS OF DIFFERENT ANOMALY
SIMULATION METHODS, ACC REPRESENTS THE PROBABILITY OF THE
GENERATED SAMPLES BEING CORRECTLY DISTINGUISHED
FROM REAL SAMPLES

Method	[23]	[1]	[73]	[74]
Acc	100.0	100.0	100.0	99.2

results, its combination with focal loss performed suboptimally compared to our default entry. Thanks to ℓ_1 's supervision of defect edges and focal loss's balance of small defects, the accuracy of our method was greatly improved in the AD task.

G. Other Discussions

1) *False Detection*: To further validate the efficacy of our proposed method in significantly reducing FPs and FNs, we conducted a thorough evaluation of the MVTec AD dataset. As shown in Table XI, our method outperforms other approaches across all four key metrics. The F_1 -score, being the harmonic mean of precision and recall, reflects a balanced enhancement in both these metrics. Specifically, our method attains an F_1 -score of 71.8%, surpassing the nearest competitor's score of 66.7%. With a precision of 71.3%, our approach ensures that a high proportion of detected anomalies are genuine, thereby significantly reducing FP rates. Simultaneously, a recall of 72.7% indicates that our method successfully identifies a larger fraction of actual anomalies, drastically lowering FN rates.

2) *Domain Gap*: To highlight the significant gap between current anomaly simulation methods and real-world anomalies, we conducted an experiment using a simple classification network. As shown in Table XII, this classifier rapidly learned

to distinguish between simulated and real anomalous after just a few training epochs. This result underscores the limitations of existing simulation techniques in capturing the complexity and variability of real-world anomalous, leading to models that perform well on synthetic data but fail to generalize to practical applications, thereby compromising their utility. However, by transforming the feature in the latent space, we can develop more realistic training data, enhancing the simulation's ability to replicate the intrinsic characteristics of actual anomalies. Our approach not only increases the diversity of the simulated data but also leads to a more robust and reliable AD model.

V. CONCLUSION

A novel denoising autoencoder with features is tailored for industrial AD tasks. Recent methods of AD that rely on reconstruction, often falter in the face of genuine anomalies during the testing phase, attributed to the scarcity of authentic defect samples throughout the training stage of the task. By ingeniously incorporating noise at the latent space, with a diverse array of reconstruction targets and a unique feature alignment module, our FeatDAE adeptly overcomes the prevalent overgeneralization and overfitting issues in reconstruction-based models. The experiments conducted on the surface AD benchmarks reveal that each of our proposed components markedly enhances the performance. Our results outperform the previous state-of-the-art methods by 5.7% pixel-wise AP on the MVTec AD dataset and 3.8% pixel-wise AP on the VisA dataset, with an extremely high inference speed. At present, the localization capability of our model exhibits variability when confronted with different classes. By taking into account disparities in size, texture, and labeling, we leave achieving uniform performance across the spectrum of anomalous and emerges as a paramount objective for future work.

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